Machine Learning in Applications
AM02 Project Report

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***Abstract*—In this paper we investigate the use of Multiple Instance Learning (MIL) on the task of cancer diagnosis on Whole Slide Images (WSIs). In particular we analyze an attention-based MIL model and compare it to traditional MIL approaches based on simple aggregation strategies. Furthermore we evaluate the performance of these models in a segmentation setting by projecting their score maps onto the spatial layout of the slides. Experimental results show that while the traditional max pooling solution provides better classification accuracy, the attention-based MIL model achieves superior discriminative performance, producing more localized and informative score maps.**

I. Introduction

In the field of cancer diagnosis, histopathological images of biopsies are widely used. These biopsies are digitized into Whole Slide Images (WSIs), which are high-resolution digital representations of entire microscope slides. WSI files are exceptionally large and are typically stored in a pyramidal format with multiple resolution levels. Such images are commonly annotated by medical professionals; however, these annotations are often provided only at the slide level or as coarse regions of interest (ROIs), rather than at the level of individual image patches. This makes it difficult to directly apply traditional supervised machine learning methods for classification.

Multiple Instance Learning (MIL) is a method devised to work in Weakly Supervised scenarios such as these. MIL deals with a bag of instances for which a single class label is assigned. The main goal of MIL is to learn a model that predicts a bag label from multiple instances. An additional challenge is to identify the most informative instances, i.e. the ones that trigger the bag label. This is very interesting in medical applications, where model interpretability is crucial to support clinical decision-making.

In this work, we investigate the application of an attention-based Multiple Instance Learning approach for the identification of tumoral tissue in histopathological slides.

II. Background

A. Multiple Instance Learning

In the classical supervised learning problem one aims at finding a model that predicts a label y for a given instance $\mathbf{x}$. In Multiple Instance Learning, one (binary) label Y is given for a bag of instances $X = \{\mathbf{x_1}, \dots, \mathbf{x_K}\}$, where instances have no ordering or dependency between one another. We assume that there exists one label y_i for each instance, however this knowledge is not available to us at training time. We assume that the bag label corresponds to

$$Y = \max_{k=1,\dots,K} \{y_k\}$$

i.e. one bag is positive if at least one of its instances is positive.

Using the maximum operator during training is not advisable, as it would introduce issues with vanishing gradients.

Instead, a model is trained by optimizing the log-likelihood:

- Each instance is fed to a feature extractor to obtain a low-dimensional embedding;
- A pooling function is used to obtain a bag representation, by aggregating the embeddings;
- A classifier is used to derive the score $\theta(X)$ from the bag representation.

The score $\theta(X)$ describes the probability of $Y = 1$ for a given bag X.

B. Attention-based MIL

In traditional MIL, fixed (non-trainable) pooling operators such as max and mean are commonly employed. However, their lack of adaptability can limit model performance.

Ilse et al. [1] propose a method based on a trained weighted average, where the weights are computed by a two-layered neural network. Let $H = \{\mathbf{h_0}, \dots, \mathbf{h_K}\}$ be a bag of K embeddings. the pooling operator is defined as such:

$$\mathbf{z} = \sum_{k=1}^{K} a_k \mathbf{h_k};$$

where the weights a_k are:

$$a_k = \frac{\exp(\mathbf{w}^T \tanh(\mathbf{V}\mathbf{h}_k^T))}{\sum_{j=1}^{K} \exp(\mathbf{w}^T \tanh(\mathbf{V}\mathbf{h}_j^T))}$$

where $\mathbf{w} \in R^{L\times 1}$ and $\mathbf{V} \in R^{L\times M}$ are parameters.

In principle this method should allow the model to learn to give different weights to different instances within a bag and, therefore, be able to detect key instances. High scores should be assigned to those instances which are likely to be positive (label $y_k = 1$) therefore the resulting model should give interpretable results.

III. Materials and methods

A. Setup

To evaluate the effectiveness of the attention-based MIL approach we worked on a dataset provided to us by Politecnico di Torino, consisting of 24 whole-slide images collected from patients diagnosed with colon adenocarcinoma. The first 14 slides were used for model training, while the remaining 10 were reserved for performance evaluation. All models were trained for 100 epochs using the Adam optimization algorithm and binary cross-entropy loss, with a learning rate of 1×10^{-4}. We carried out the experiments using a personal computer, equipped with a NVIDIA GeForce RTX 5090. The implementation is available in our GitHub repository: https://github.com/GNNatan/MLA-Project.

B. Preprocessing

We first performed preprocessing on the Whole Slide Images. Each slide was divided into non-overlapping patches of size 256×256 pixels. A simple thresholding strategy was applied to discard low-information patches that mainly correspond to background regions. The remaining patches were resized to 224×224 pixels and normalized before being fed to the model.

C. Feature extraction

To transform the slices into a low-dimensional embedding, we used the model proposed by Sirinukunwattana et al. [2]. The feature extraction pipeline is composed of two sequential modules. The first module consists of two convolutional layers with ReLU activations and intermediate max-pooling operations. The feature maps are then passed through the second module, where they are flattened before going through two fully-connected layers, with ReLU activation and Dropout regularization.

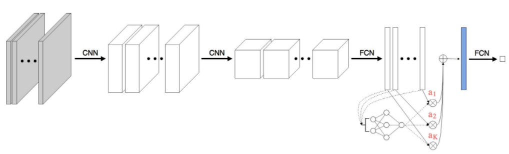

Fig. 1. Model architecture for the implementation of attention-based MIL. Figure taken from [1]. After extracting instance-level features, an attention mechanism assigns a weight to each instance. A bag-level representation is then obtained as a weighted aggregation of the instance embeddings, and classification is performed on this aggregated representation.

D. Pooling

We evaluated the model using three different pooling layers. Let $H = \{\mathbf{h_0}, \dots, \mathbf{h_K}\}$ be a bag of K embeddings:

- The *maximum pooling* operator emulates a classical instance-level decision rule, where the highest scoring instance determines the bag prediction, it is defined as:
$$\forall_{m=1,\dots,M} : z_m = \max_{k=1,\dots,K} \{\mathbf{h}_{km}\};$$
- The *mean pooling* operator computes the bag prediction from the mean of the instance embeddings, the operator is defined as:
$$\mathbf{z} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{h}_k;$$
- The *attention-based pooling* operator, following the formulation in Section II-B, computes the bag representation as a weighted sum of the instance embeddings, where the weights are learned through a trainable attention mechanism.

E. Classification

To obtain the final classification score, we used a simple classifier based on Logistic Regression.

F. Addressing bag imbalance

The adopted MIL formulation defines a bag as positive if it contains at least one positive instance, and negative if it contains only negative instances. Due to this formulation, bag-level labels are inherently imbalanced, with positive bags being substantially more frequent.

To mitigate this imbalance and promote stable learning during training, a balancing strategy was integrated directly into the bag construction procedure. Bags are generated iteratively from the pool of available instances, while tracking the current number of positive and negative bags. Whenever the number of positive bags exceeds that of negative bags, the subsequent bag is generated under a negative constraint, i.e. by sampling exclusively from negative instances. Otherwise, the bag is generated without constraints. This procedure is repeated until all instances have been assigned to a bag.

IV. Results and discussion

A. Classification Performance

	Performance metrics				
Model name	Accuracy	Precision	Recall	F-Score	AUC
dummy	0.909	0.909	**1.0**	0.952	0.5
attention	0.835	**1.0**	0.823	0.903	**0.999**
mean	0.580	**1.0**	0.548	0.708	0.946
max	**0.966**	0.989	0.974	**0.982**	0.976

TABLE I
BAG-LEVEL CLASSIFICATION PERFORMANCE FOR THE DIFFERENT MODELS, TESTED ON SLIDES 14 TO 24

To evaluate the performance of the different pooling methods we compared them with each other, as well as with a *dummy* model, that always predicts the positive class for each bag.

The performance of the models is summarized in Table I. As expected, the *dummy* baseline achieves high accuracy due to the strong class imbalance in the dataset, but its low AUC showcases its lack of discriminative power.

Among the MIL-based approaches, the *attention* model achieves the highest AUC, indicating a strong ability of ranking positive and negative samples, however its accuracy falls short of the *max* model suggesting that the predicted scores might not be well calibrated with respect to the decision threshold.

The *max* model achieves the best overall classification performance, with the highest accuracy and F1-score, while mantaining a high AUC. This suggests that the traditional MIL assumption, in which a bag is considered positive if at least one instance is positive, is particularly fitting for this dataset.

The *mean* model performs worse than both *attention* and *max*, likely due to its tendency to average instance-level features, which can dilute the contribution of discriminative patches.

It is important to note that the ground truth annotations used in this work are inherently weak and approximate. Bag labels are derived from coarse regions of interest rather than precise instance-level annotations, therefore the evaluation metrics computed at the bag level may not fully reflect the capabilities of the models. In this setting, models that are more sensitive to subtle patterns, such as attention-based approaches, may assign low scores to less informative areas of the annotated regions, these predictions are counted as errors, potentially leading to an underestimation of the model performance.

B. Segmentation Performance

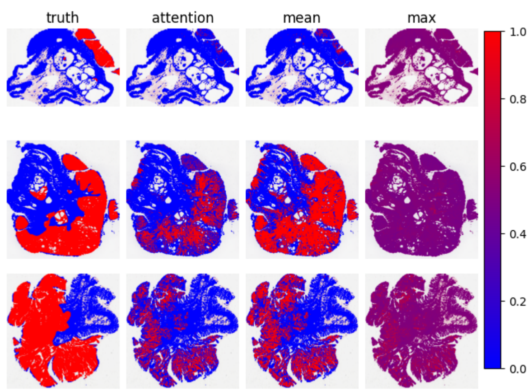

Fig. 2. Ground truth and predictions from the three models on slides 22, 23 and 24. Each pixel corresponds to a patch, and the color represents the predicted probability of tumor presence, ranging from blue (low) to red (high).

To get a better understanding of the capabilities of the models, we generated score maps by projecting the single-instance bag scores back into the spatial layout of the slide (Figure 2).

From a qualitative analysis, we can see that the *attention* model seems to better discriminate between positive and negative instances, assigning higher scores to samples within the human-annotated regions.

The *mean* model achieves similar results, although with a less pronounced separation between the two classes.

The *max* model, while achieving strong bag-level classification performance, produces less informative score maps. In particular, it tends to assign high scores to both positive and negative patches, resulting in reduced spatial specificity.

V. Conclusions and Future Works

In this work, we investigated the application of Multiple Instance Learning (MIL) for cancer diagnosis on Whole Slide Images (WSIs) under a weakly supervised setting. We compared different aggregation strategies: max pooling, mean pooling, and the attention-based approach. We showed that, despite the strong class imbalance, max pooling provides the best classification results while the attention-based model offers superior discriminative capabilities producing more interpretable score maps. This highlights the potential of attention-based MIL approaches in medical imaging applications, where model interpretability and localization capabilities are of particular importance.

Several directions for future work may further improve model performance. In particular, the adoption of data augmentation strategies could enhance generalization in low-data regimes such as the one considered in this study. Additionally, alternative bag balancing techniques could be explored to better address class imbalance at the bag level. Finally, investigating different attention mechanisms within the pooling layer may provide further improvements in both predictive performance and interpretability.

References

[1] M. Ilse, J. M. Tomczak, and M. Welling, “Attention-based deep multiple instance learning,” *arXiv preprint arXiv:1802.04712*, 2018.

[2] K. Sirinukunwattana, S. E. A. Raza, Y.-W. Tsang, D. R. J. Snead, I. A. Cree, and N. M. Rajpoot, “Locality sensitive deep learning for detection and classification of nuclei in routine colon cancer histology images,” *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1196–1206, 2016.